

RODOLPHE ABOU ASSALI

Nationality: Lebanese

Born on August 4, 2000

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[R^G](#), [ArXiv](#), [HAL](#)

Education

- **2023–Present:** Joint PhD in Differential Geometry (3rd year)
Topic: The biharmonic Steklov problem for differential forms
Supervisors: [Nicolas Ginoux](#), [Georges Habib](#), [Samuel Tapie](#)
Lebanese University, Faculty of Science, Lebanon – University of Lorraine, [Institut Élie Cartan](#), France.
- **2022–2023:** Master 2 in Fundamental and Applied Mathematics (Double degree)
Lebanese University, Faculty of Science, Lebanon – University of Lorraine, Institut Élie Cartan, France.
- **2021–2022:** Master 1 in Mathematics
Lebanese University, Faculty of Science, Section II, Lebanon.
- **2018–2022:** University Diploma in Education
Lebanese University, Faculty of Science, Section II, Lebanon.
- **2018–2021:** Bachelor's Degree in Mathematics
Lebanese University, Faculty of Science, Section II, Fanar, Lebanon.

Professional Experience

- **2025–2026:** Lecturer at IUT Nancy – Charlemagne – Computer Science Department
Courses taught:
 - Descriptive Statistics (L1)
 - Numerical methods (L1)
 - Graph theory (L1)
 - Optimization Methods (L2, DACS).
- **2024–2025:** Lecturer at IUT Nancy – Charlemagne – Computer Science Department
Courses taught:
 - Descriptive Statistics (L1)
 - Optimization Methods (L2, DACS)
 - Mathematical Complements and Modelling (L3, RA–IL).

- **2022–2023:** Master 2 Internship in Differential Geometry
Topic: Spectral Theory of the Biharmonic Steklov Operator
Supervisors: [Nicolas Ginoux](#), [Georges Habib](#), [Samuel Tapie](#)
 Lebanese University, Faculty of Science, Lebanon – University of Lorraine, [Institut Élie Cartan](#), France.
- **2018–2022:** Private tutoring – Mathematics and Physics for middle and high school students.

Excellence Scholarships and Research Grants

- **2023–2026:** PhD scholarship from the [SAFAR](#) program of the French Embassy in Lebanon, for a PhD under the supervision of Nicolas Ginoux, Georges Habib, Samuel Tapie.
- **2026:** Orion Grant for Participation in International Conferences.
- **2025:** [DrEAM](#) Grant from the University of Lorraine under the Lorraine University of Excellence (LUE) initiative, for a two-month research stay at Sapienza University under the supervision of [Alessandro Savo](#).
- **2023:** Master Grant from the [Orion](#) program at the University of Lorraine for a 4-month research stay at the [IECL](#) under the supervision of [Nicolas Ginoux](#), [Georges Habib](#), [Samuel Tapie](#).

Preprints

- **2026:** *Geometric eigenvalue estimates of Kuttler-Sigillito type on differential forms.* [arXiv:2602.09876 \[math.DG\]](#).
- **2025:** *Biharmonic Steklov problems with Neumann boundary conditions and spectral inequalities on differential forms.* [arXiv:2507.05049 \[math.DG\]](#).

Talks

Speaker

- **2026:** [Phd away days](#) at University of Luxembourg
Steklov problems and spectral inequalities in planar domains.
- **2025:** [Seminar of the department of Basic and Applied Sciences for Engineering](#) at Sapienza University, Rome
Biharmonic Steklov problems and spectral inequalities for differential forms.
- **2025:** [Advanced PDE Seminar](#), Department of Mathematics and Statistics, University of Konstanz, Germany
Biharmonic Steklov problems and spectral inequalities for differential forms.
- **2025:** [Differential Geometry Seminar](#), IECL Nancy
Biharmonic Steklov problems and spectral inequalities on differential forms.

- **2024:** [IECL PhD Students Day](#), Nancy
The Biharmonic Steklov Operator.
- **2024:** [Doctoral Forum of the EDST](#), Lebanese University, Beirut
The Biharmonic Steklov Problem on Differential Forms.

Attendee

- **2025:** Differential Geometry at L'Aquila ([DG@AQ25](#)).
- **2024:** European Doctorate School of Differential Geometry ([ESDG](#)), Granada.
- **2024:** Working group on the spectrum of the Steklov operator, IECL Nancy.
- **2023:** Geometry Day, IECL Nancy.

Skills

Computer Skills:

- **Programming languages:** C, CSS, C++, HTML, MATLAB (Beginner), $\text{\LaTeX}$.
- **Software:** Proteus 8 Professional, Tinkercad, Cisco Packet Tracer, Oracle Data Modeler (Intermediate).
- Adobe Photoshop, Illustrator, Dreamweaver, Visual Studio, VMware Workstation (Beginner).

Language Skills:

- Arabic (Native language)
- French (Advanced level)
- English (Advanced level)

Activities

- **2026-Present:** Elected PhD student representative on the IECL Council.
- **2026-Present:** Member at Société de Mathématiques Appliquées et Industrielles, ([SMAI](#)).
- **2026:** Co-organizer of the [PhD away days](#) in Luxembourg with the PhD students from the University of Luxembourg.
- **2025-Present:** Member at Société Mathématique de France, ([SMF](#)).
- **2024-Present:** Co-organizer of the [PhD Students Seminar](#) at IECL.
- **2023-Present:** University teaching training sessions – Université de Lorraine.
- **2025:** Co-organizer of the [PhD away days](#) in Nancy with the PhD students of the University of Luxembourg.
- **2025:** Workplace First Aid Certificate (SST) – Nancy.

- **2024–2025:** Co-facilitator of the [Math Club](#) at Lycée Jacques Callot, Vandœuvre-lès-Nancy.
- **2024–2025:** Co-facilitator for Math workshops at [the Science Festival](#) at FST and Jeanne d’Arc Elementary School.
- **2021–2023:** Student representative in M1 and M2.